PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

Table of Contents

1. Project Overview 2
2. Background & Business Objective 2
3. Product Purpose 3
4. System Functionalities 3
5. User Stories & Use Cases 4
6. Features Included (Scope Definition) 4
7. Features Not Included (Scope Control) 4
8. Assumptions & Constraints 5
9. Risks & Questions Throughout Development 5

Project Name:TernakAI

Project Name: TernakAI
Version: 1.0

1. Project Overview

- **Problem Statement:** Smallholder poultry farmers managing 1,000–10,000 birds lack unified early-warning tools, forcing them to rely on reactive intuition. By the time diseases like CRD or Avian Influenza present visible symptoms, the outbreak is 48–72 hours advanced, locking in 30–50% mortality. The historical cost of this delayed detection is devastating: Malaysia's poultry industry has lost MYR 1.5 billion to bird flu, exemplified by the 2017 Kelantan outbreak where over 33,000 birds were systematically culled (*Poultry World*). Because feed consumes up to 75% of their variable costs, smallholders cannot absorb these shocks; late-stage detection and the resulting mass culling routinely push them out of business (Wan Mohd Noor et al., 2025).
- **Target Domain:** Agritech / Predictive AI Decision Support for Livestock Farming.
- **Proposed Solution Summary:** TernakAI removes the analytical blind spots by fusing structured farm data (telemetry) with unstructured data (farmer notes). It utilizes deterministic baseline algorithms combined with Z.AI's GLM to act as an early-warning reasoning engine. It shifts the farmer's role from reactive treatment to proactive disease prevention by providing instant diagnoses, prioritized action plans, and projected financial impacts.

2. Background & Business Objective

- **Background of the Problem:** Traditional poultry management operates on a fragmented, manual basis. A farmer might note a slight drop in feed, later feel the shed is warm, and notice the birds are unusually quiet. Because these isolated signals are not synthesized, the compound risk is ignored. Research in precision poultry farming confirms that this reliance on fragmented metrics is fundamentally insufficient for modern flock management (Paneru et al., 2025). This lack of data synthesis prevents early intervention, meaning that multi-signal disease patterns go unnoticed. By the time visual symptoms trigger a call to the veterinarian, the disease has already spread throughout the flock.
- **Importance of Solving This Issue:** By accelerating detection from days to hours, the system can reduce outbreak mortality from 30–50% to 5–10%. For a typical 5,000-bird farm, this translates to saving RM7,000–RM14,000 per incident. It breaks the barrier of requiring expensive on-site veterinary intuition for early-stage anomaly detection.

- **Strategic Fit / Impact:** It aligns with national food security and agricultural modernization initiatives. The integration of Z.AI GLM empowers non-technical rural farmers with enterprise-grade decision intelligence using standard mobile devices.

3. Product Purpose

- **3.1. Main Goal of the System:** To provide an AI-enabled, context-aware decision intelligence partner for poultry disease prevention.
- **3.2. Intended Users (Target Audience):**
    - Smallholder Poultry Farmers (1,000 - 10,000 birds)
    - Farm Supervisors / Managers
    - Contract Farming Operators

4. System Functionalities

- **4.1. Description:** The system operates as a high-speed reasoning engine for farm anomalies. Using structured data and natural language processing, it captures signal deviations, cross-references them against veterinary protocols, and outputs constraint-aware medical and operational recommendations.
- **4.2. Key Functionalities:**
    - **Data & Signal Fusion:** Translates manual daily inputs (feed, temp, mortality) and free-text notes (Malay/English) into deterministic risk scores.
    - **Insight & Action Engine:** Generates time-bound, prioritized mitigation strategies (e.g., immediate ventilation changes, 12-hour supplement additions).
    - **Scenario Narration:** Projects expected final mortality and financial loss (in RM) comparing 'Action' vs. 'No Action' scenarios.
- **4.3. AI Model & Prompt Design**
    - **4.3.1. Model Selection:** TernakAI uses Z.AI GLM as an optional reasoning layer on top of a deterministic poultry-risk engine. The backend first computes a transparent RiskAssessment from farm data such as feed intake, temperature, mortality, ventilation condition, behaviour flags, and farmer notes. GLM is then used to convert that structured assessment into four outputs that deterministic scoring alone cannot provide well: plain-language interpretation, ranked disease

hypotheses, prioritised actions,
and scenario narration. This
design is deliberate: the core scoring system remains usable without the model, while GLM adds the reasoning and communication layer needed for farmer decision support.

- **4.3.2. Prompting Strategy:** The system uses a **two-stage grounded prompting flow**. First, it retrieves disease context from a local poultry knowledge base containing DVS/VRI, MARDI, and related veterinary reference chunks. Second, it sends that grounded context into **two separate GLM calls**: a **reasoning call** for interpretation and hypothesis ranking, followed by an **action-planning call** for recommendations and "what happens next" narration. The prompts are heavily structured and require fixed output sections such as `INTERPRETATION`, `HYPOTHESIS_1-3`, `ACTION_1-3`, `NO_ACTION`, `ACT_12H`, and `ACT_48H`. This reduces free-form drift and keeps model outputs anchored to retrieved evidence and deterministic farm signals.
- **4.3.3. Context & Input Handling:** The model does not receive raw, unconstrained chat logs. It is fed a compact farm-state summary built by the backend: current risk score and trend, deviations from baseline, flock metadata, recent trajectory, and short farmer notes. In the API schema, farmer notes are capped at 500 characters, while structured fields such as ventilation condition and behaviour flags are validated against fixed enums. The system also supports English, Bahasa Melayu, and bilingual ("bahasa rojak") prompting modes, allowing short field notes like "ayam senyap sikit, kurang makan" to be incorporated directly into analysis.
- **4.3.4. Fallback & Failure Behavior:** If the live GLM call fails, returns unusable structure, or the provider is unavailable, TernakAI does **not** block the workflow. Instead, it falls back to an **offline local analysis path** that still returns interpretation, heuristic disease hypotheses, recommendations, and scenario narration using deterministic logic plus retrieved knowledge-base context. Outputs are explicitly labeled as either `z-ai-glm` or `offline-glm-fallback`, so the frontend can distinguish live model reasoning from local fallback behaviour. This architecture is important for field reliability: farmers still receive a working risk assessment and action guidance even when the external model layer is down.

5. User Stories & Use Cases

- **User Stories:** "As a smallholder farmer, I want to input my daily feed usage and quick observations in Malay, so that the AI can warn me if my birds are getting sick before I can physically see it."
- **Use Cases (Main Interactions):**

- **Early Detection Flow:** Farmer inputs daily metrics -> System detects a 3-day negative deviation in feed + flags the keyword "quiet" -> System calculates a Moderate Risk Score and alerts the farmer to monitor ventilation.
- **Diagnostic & Action Flow:** System detects High Risk -> Retrieves CRD protocol via RAG -> GLM outputs diagnosis hypothesis, specific actionable steps (e.g., isolate Shed A), and projects RM losses if ignored.

6. Features Included (Scope Definition)

- **Data Ingestion & Integration:** Manual daily input via mobile form including shed temperature (°C), feed intake (kg), water intake (litres), mortality count, ventilation condition (4-level structured selection: normal / mild / strong / sensor_high), behaviour checklist flags (water_change, abnormal_sounds, huddling_panting, reduced_movement), and free-text farmer notes (Malay/English). Multiple readings per day are supported; the system aggregates them into a single daily summary.
- **Deterministic Logic Engines:** Baseline & Deviation Engine (3-day rolling average, age-normalized using per-day growth/temperature-drop factors) and Risk Scoring Engine mapping compound signals to risk thresholds across 6 categories plus two additive bonuses (combination and sustained-trend), capped at 100.
- **Z.AI GLM Decision Intelligence:** Interpretation Engine, RAG-Grounded Disease Hypothesis Generator (using DVS/OIE knowledge base via ChromaDB with a custom embedding function), Insight Generation, and Constraint-Based Action Recommendation Engine.
- **User Interface — Today Tab:** Risk score, signal cards, and score breakdown. Compares today's readings against the flock's rolling baseline.
- **User Interface — Trends Tab:** Multi-variable timeline with configurable analysis windows (3 days, 1 week, 2 weeks), graph filters, and AI Analysis Hub generating overall assessment, disease probability, recommended actions, and 7-day impact comparison for the selected window.
- **User Interface — History Tab:** Chronological log of all past daily reports with per-day risk score, top driver, and full report drill-down. Provides audit traceability.
- **Smart Alerts:** Triggered reactively on reading submission when the calculated risk score meets critical thresholds. Deduplication logic prevents duplicate alert entries.
- **Bilingual Support:** GLM prompts and outputs default to bilingual (Malay/English), configurable to `en` or `ms`.

7. Features Not Included (Scope Control)

- **Hardware Integration:** IoT sensor integrations (temperature, humidity, ammonia) planned for Phase 2 post-hackathon scaling.
- **Push / SMS Notifications:** Real-time alert delivery to the farmer's phone outside the app when critical risk thresholds are breached.
- **Multi-Flock Management:** Managing multiple batches or farms under one account. Currently hardcoded to a single flock (flock_2026_batch3).
- **Data Export (PDF/CSV):** Generating downloadable daily or weekly reports for record-keeping or sharing with veterinarians.
- **Veterinary Portal:** A separate dashboard view for vets or DVS officers to remotely review flagged farms and respond to alerts.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Estimated token cost is extremely low per session (~1,000 tokens for input/RAG context, ~500 tokens for output). To keep inference costs manageable at scale, the system uses **baseline caching**—only sending the mathematical deviations to the GLM rather than the entire raw history of the farm, and caching common risk-pattern outputs.
- **Technical Constraints:** The MVP vector database (RAG) will run locally via FAISS/ChromaDB. Advanced continuous data-streaming pipelines are not supported in the hackathon build.
- **Performance Constraints:** Inference must happen within seconds over 4G/3G mobile networks, dictating that the UI remains lightweight without heavy visual assets.
- **User Input:** The model relies on the farmer logging data at least once every 24 hours. The AI cannot project accurately if the data gap exceeds 2 days.

9. Risks & Questions Throughout Development

- **User Friction:** Manual data entry can be tedious. *How do we ensure farmers consistently use the app daily without abandoning it?* (Mitigation: Limit inputs to 4 fields and optimize mobile UI).

- **Medical Hallucination:** *How do we guarantee the AI does not prescribe lethal or incorrect medication dosages?* (Mitigation: GLM temperature set to 0.0; all medical outputs strictly forced to cite RAG documents).
- **Data Cold Start:** *How does the system calculate risk on Day 1 when there is no historical baseline?* (Mitigation: Implement a 3-day calibration phase where AI output is restricted to general observation).